



DevOps

Important Commands

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● Page 1 – Linux & Git Essentials



Linux Essentials



Powerful commands to navigate, manage and operate your system.

```
01 pwd # Current directory
02 ls -la # List all files (long format)
03 cd folder # Change directory
04 mkdir project # Create directory
05 touch file.txt # Create file
06 cp source dest # Copy file or directory
07 mv old new # Move or rename
08 rm -rf folder # Remove directory (force)
09 find . -name "*.js" # Find files by name
10 grep "text" file # Search text in file
11 chmod +x script.sh # Make file executable
12 ps -ef # Show running processes
13 top # Real-time system monitor
14 df -h # Disk space usage
15 free -m # Memory usage
```

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Git Essentials



Track changes, collaborate and manage your code like a pro.

```
01 git init # Initialize a new repository
02 git clone <repo> # Clone a remote repository
03 git status # Check status of repository
04 git add . # Stage all changes
05 git commit -m "message" # Commit changes
06 git push origin main # Push to remote repo
07 git pull origin main # Pull latest changes
08 git checkout -b feature # Create & switch branch
09 git merge feature # Merge branch into current
10 git stash # Save changes temporarily
11 git log # View commit history
```


DevOps Important Commands

● Page 5 – Helm, Monitoring & Networking



Helm

Package Manager for Kubernetes



```
01 helm repo add
02 helm repo update
03 helm install app chart
04 helm upgrade app chart
05 helm list
06 helm uninstall app
```



Monitoring

Observe & Analyze Systems



```
01 systemctl start prometheus
02 systemctl start grafana-server
03 systemctl status prometheus
04 systemctl status grafana-server
```

 PYTHON LIFE



Prometheus
Metrics Collection



Grafana
Visualization & Dashboards



Networking & SSH

Connect, Transfer & Troubleshoot



```
01 ping google.com           # Check connectivity
02 curl https://example.com   # Fetch data
03 wget URL                   # Download file
04 netstat -tulnp             # Network connections
05 ss -tulnp                  # Socket statistics
06 nslookup google.com        # DNS lookup
07 dig google.com             # Detailed DNS lookup
-----
08 ssh user@server            # Connect to server
09 scp file.txt user@server:/home/ # Secure file copy
10 ssh-keygen                 # Generate SSH key
```



Must-Know Interview Commands

```
01 git status
02 docker ps
03 docker logs
04 kubectl get pods
05 kubectl apply -f
06 terraform plan
07 terraform apply
08 ansible-playbook
09 helm install
10 aws configure
11 curl
12 ssh
```



DevOps Success Path



Master Git → Docker → Kubernetes → Jenkins → AWS → Terraform
to build a strong DevOps career. 🚀



DevOps

Important Commands

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• Page 4 – Cloud & Infrastructure as Code



AWS CLI

Cloud Management

Manage and automate AWS services from the command line.



```
01 aws configure # Configure AWS CLI
02 aws s3 ls # List S3 buckets
03 aws s3 cp file.txt s3://bucket/ # Copy file to S3
04 aws ec2 describe-instances # List EC2 instances
05 aws iam list-users # List IAM users
06 aws lambda list-functions # List Lambda functions
```

PYTHON LIFE



Terraform

Infrastructure as Code

Provision and manage infrastructure using code, consistently and repeatably.



```
01 terraform init # Initialize working directory
02 terraform validate # Validate configuration
03 terraform fmt # Format code
04 terraform plan # Preview changes
05 terraform apply # Apply changes
06 terraform destroy # Destroy infrastructure
07 terraform output # Show output values
```



Ansible

Configuration Management

Automate software configuration, deployment and task orchestration.



```
01 ansible all -m ping # Check connectivity
02 ansible-playbook playbook.yml # Run playbook
03 ansible-galaxy install role # Install role from Galaxy
```

Infrastructure Automation Flow





DevOps Important Commands

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● Page 3 – CI/CD & Build Tools



Jenkins

Automation Server

Automate builds, tests and deployment pipelines with ease.



```
01 systemctl start jenkins
02 systemctl stop jenkins
03 systemctl restart jenkins
04 systemctl status jenkins
```

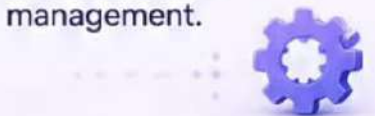
PYTHON LIFE



Maven

Build Automation

Build, test and manage Java projects with powerful lifecycle management.



```
01 mvn clean
02 mvn compile
03 mvn test
04 mvn package
05 mvn install
06 mvn spring-boot:run
```



Gradle

Flexible Build Tool

Fast, flexible and extensible build automation tool.



```
01 gradle clean
02 gradle build
03 gradle test
04 gradle run
```



CI/CD Pipeline Flow





DevOps Important Commands

Page 2

● Page 2 – Docker & Kubernetes



Docker



Docker helps you build, share and run applications in isolated containers.

```
01 docker images           # List Docker images
02 docker ps               # List running containers
03 docker ps -a            # List all containers
04 docker pull nginx        # Pull nginx image
05 docker run -d -p 80:80 nginx # Run nginx container
06 docker stop container    # Stop container
07 docker restart container # Restart container
08 docker exec -it container bash # Exec into container
09 docker logs container    # View container logs
10 docker build -t app .    # Build Docker image
11 docker-compose up -d     # Start services in background
12 docker-compose down      # Stop and remove services
```



Kubernetes



Kubernetes helps you deploy, manage and scale containerized applications.

```
01 kubectl get nodes       # List cluster nodes
02 kubectl get pods         # List pods
03 kubectl get deployments  # List deployments
04 kubectl get services     # List services
05 kubectl describe pod pod-name # Describe a pod
06 kubectl logs pod-name    # View pod logs
07 kubectl exec -it pod-name -- bash # Exec into pod
08 kubectl apply -f deployment.yaml # Apply configuration
09 kubectl delete -f deployment.yaml # Delete configuration
10 kubectl rollout restart deployment/app # Restart deployment
```



Pro Tip

Learn Docker before Kubernetes.

